

HYDROGENATED NBR: PATENT LANDSCAPE ON HYDROGENATION, METATHESIS-ENABLED MOLECULAR WEIGHT CONTROL, AND HNBR-CONTAINING FORMULATIONS

1. ABSTRACT

This report synthesizes a 2010–2024 patent set relevant to hydrogenated nitrile butadiene rubber (HNBR) and closely related HNBR-containing formulations. The target is defined as nitrile rubber precursors that are hydrogenated, partially hydrogenated, or highly saturated to reduce residual unsaturation and improve heat, ozone, and oil resistance while preserving processability and molecular weight retention. Across the selected patents, two major technical directions emerge. First, several patents focus on direct or indirect synthesis of HNBR from NBR by hydrogenation, including latex hydrogenation, solution/cement hydrogenation, and hydrogenation after metathesis-based chain scission to lower Mooney viscosity before saturation. These disclosures emphasize catalyst choice, hydrogen pressure, temperature, and the need to control residual double bond content and molecular weight distribution. Second, multiple patents use HNBR as a functional ingredient in downstream compositions rather than as the primary synthesis target, including swellable packers, electroconductive silver pastes, toothed belts, and battery electrode dispersants. These formulation patents are still relevant because they specify HNBR grades, hydrogenation levels, and processing windows that indirectly define useful HNBR property envelopes.

The landscape shows recurring use of ruthenium-based metathesis catalysts, often with salt additives or alkaline earth chlorides, to depolymerize NBR prior to hydrogenation and thereby obtain lower-Mooney HNBR with narrower molecular weight distribution. Other patents disclose conventional emulsion polymerization of NBR followed by palladium-catalyzed hydrogenation in latex or solution, with hydrogen pressures around 3 MPa and temperatures near 50°C in representative examples. In formulation-oriented patents, HNBR is selected for improved thermal stability, low residual unsaturation, and compatibility with fillers, SAPs, silver powders, or polyamide resins. Overall, the evidence supports HNBR synthesis routes that combine controlled NBR polymerization, selective hydrogenation, and, where needed, metathesis-assisted molecular weight reduction to balance processability with final elastomer performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2692788B1
- Patent Title: Highly saturated nitrile rubber composition and crosslinked rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a formulation patent for highly saturated nitrile rubber compositions rather than a new hydrogenation route. It discloses preparing carboxyl group-containing highly saturated nitrile rubber and non-carboxylated highly saturated nitrile rubber by emulsion polymerization of NBR precursors followed by palladium-catalyzed hydrogenation in latex, then compounding them with polyamide resin and additives using a twin screw extruder and Banbury mixer. - Acrylonitrile: 37 parts — Synthesis Example 1 - Mono n-butyl maleate: 6 parts — Synthesis Example 1 - t-Dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-Butadiene: 57 parts — Synthesis Example 1 - Reaction temperature: 5°C — Synthesis Example 1 - Polymerization time: 16 hours — Synthesis Example 1 - Hydrogen pressure: 3 MPa — Synthesis Example 1 - Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Palladium content: 1,000 ppm with respect to dry rubber — Synthesis Example 1 - Methanol: 2 volumes — Synthesis Example 1 - Vacuum drying: 60°C for 12 hours — Synthesis Example 1 - Acrylonitrile: 42 parts — Synthesis Example 2 - 1,3-Butadiene: 58 parts — Synthesis Example 2 - t-Dodecyl mercaptan: 0.45 part — Synthesis Example 2 - Reaction temperature: 5°C — Synthesis Example 2 - Polymerization time: 16 hours — Synthesis Example 2 - Hydrogen pressure: 3.0 MPa — Synthesis Example 2 - Palladium-silica catalyst: 500 weight ppm — Synthesis Example 2 - Vacuum drying: 60°C for 12 hours — Synthesis Example 2 - Acrylonitrile: 37 parts — Synthesis Example 3 - Methacrylic acid: 4 parts — Synthesis Example 3 - 1,3-Butadiene: 59 parts — Synthesis Example 3 - Reaction temperature: 5°C — Synthesis Example 3 - Polymerization time: 16 hours — Synthesis Example 3 - Carboxyl group-containing highly saturated nitrile rubber (a1): 21 parts — Example 1 - Highly saturated nitrile rubber (a2): 49 parts — Example 1 - Nylon 12: 30 parts — Example 1 - Twin screw extruder kneading temperature: 220°C — Example 1 - MT carbon black: 40 parts — Example 1 - Tri-2-ethylhexyl trimellitate: 10 parts — Example 1 - 4,4'-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - 1,3-bis(t-butylperoxyisopropyl)benzene (40% product): 7 parts — Example 1 - Twin screw extruder kneading temperature: 320°C — Example 2 - Nylon 6: 30 parts — Example 3 -

Carboxyl group-containing highly saturated nitrile rubber (a1): 10.5 parts — Example 4 - Highly saturated nitrile rubber (a2): 59.5 parts — Example 4 - Carboxyl group-containing highly saturated nitrile rubber (a1): 24 parts — Example 5 - Highly saturated nitrile rubber (a2): 56 parts — Example 5 - Nylon 12: 20 parts — Example 5

Relevant Experimental Evidence - Synthesis Example 1: carboxylated NBR was emulsion-polymerized and hydrogenated in latex at 3 MPa and 50°C to give highly saturated rubber with iodine value 7. - Synthesis Example 2: non-carboxylated NBR was polymerized and hydrogenated in acetone solution with 500 ppm palladium-silica catalyst at 3.0 MPa to give HNBR-like material with iodine value 7. - Synthesis Example 3: methacrylic-acid-containing NBR was hydrogenated to a highly saturated rubber with iodine value 9. - Example 1: a composition using 21/49 parts of carboxylated/highly saturated rubber and 30 parts Nylon 12 showed good rollability and high-temperature properties. - Example 2: increasing twin-screw kneading temperature from 220°C to 320°C was evaluated against Example 1.

Technical Relevance This patent is relevant because it discloses the precursor hydrogenation step that generates highly saturated nitrile rubbers used in HNBR-type compositions. It also provides explicit monomer ratios, hydrogenation conditions, and residual unsaturation indicators that define practical HNBR feedstocks.

Patent 2

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a synthesis patent for HNBR itself; it discloses swellable rubber compositions for downhole packers that may include a hydrogenated nitrile elastomer as one optional nitrile rubber component. The focus is on compounding nitrile elastomer/butadiene rubber with powder super absorbent polymer, curing, and swelling performance in brine and high-temperature environments. - Temperature: 330 degrees Fahrenheit — Example 1 - Pressure: 70 psi — Example 1 - Time: 2.5 hours — Example 1 - Temperature: 275 degrees Fahrenheit — Example 1 - Pressure: about 300 psi — Example 1 - Differential pressure: at least 5000 psi — Example 1 - Base pipe OD: 4.5 inches — Example 1 - Packer OD: about 5.7 inches — Example 1 - Element length: 5 feet — Example 1 - Salinity: 16% — Example 1 - Pressure requirements: 5000 psi — Example 1 - Refresh schedule: 0 hours — Example 1 - Expected contact time: 5 days — Example 1 - Time to apply pressure: 17 days — Example 1 - Powder SAP loading: 110 phr — Example 2 - Powder SAP loading: 130 phr — Example 2 - Powder SAP loading: 150 phr — Example 2 - Powder SAP loading: 170 phr —

Example 2 - Medium granular SAP mean particle size: 298.67 μm — Table 16 - Large granular SAP mean particle size: 666.23 μm — Table 16 - Powder SAP mean particle size: 35.33 μm — Table 16 - Powder SAP loading: about 205 phr — Example 3 - Powder SAP loading: about 190 phr — Example 3 - Powder SAP loading: about 160 phr — Example 3 - Powder SAP loading: at least about 150 phr — Example 2A - Powder SAP loading: at least about 175 phr — Example 2A - Powder SAP loading: at least about 197 phr — Example 2A - Powder SAP loading: at least about 200 phr — Example 2A - Powder SAP loading: at least about 205 phr — Example 2A - Powder SAP loading: at least about 225 phr — Example 2A - Heat: about 200 degrees Fahrenheit — Example 3A - Heat: about 275 degrees Fahrenheit — Example 4A - Brine: about 5% weight/volume — Example 5A - Brine: about 16% weight/volume — Example 6A - Swelling: at least about 50% — Example 7A - Swelling: at least about 75% — Example 7A - Duration: about 2 days — Example 8A - Duration: about 5 days — Example 8A - Pressure: at least about 3000 psi — Example 9A - Pressure: at least about 5000 psi — Example 9A - Pressure: at least about 8000 psi — Example 9A - Temperature: at least about 200 degrees Fahrenheit — Example 11A - Temperature: at least about 275 degrees Fahrenheit — Example 12A

Relevant Experimental Evidence - Example 1: a swellable packer cured at 330°F and 70 psi was tested in 16% brine at 275°F and withstood >5000 psi differential pressure. - Example 2: powder SAP loading from 110 to 170 phr was compared across particle sizes, showing swelling and leaching trends. - Example 3: predicted 75% swell loadings were estimated as about 205 phr for powder SAP, 190 phr for medium granular SAP, and 160 phr for large granular SAP. - Example 4: a 10% HNBR formulation showed greater swell rate than a 10% NBR formulation in several brine/temperature environments. - Example 1A: the claims define packer sealing methods using powder SAP, 200°F or 275°F heat, and 5% or 16% brine.

Technical Relevance This patent is relevant because it explicitly uses HNBR as a nitrile elastomer option in a high-temperature swellable rubber system. It helps define downstream performance windows for HNBR-containing formulations, especially where thermal stability and brine resistance are important.

Patent 3

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions of nitrile rubber, including catalyst/salt combinations that accelerate metathesis and can be used before hydrogenation to make lower-molecular-weight HNBR. The

process is a two-step route: metathetic degradation of NBR in the presence of ruthenium-based catalysts and salts, followed by hydrogenation to obtain HNBR with reduced molecular weight and narrower distribution. - Salt:additive ratio: 0.01:1 to 10000:1 — Claim 1 - Example 7 LiBr: 1 mg — Example 7 - Example 7 Grubbs II catalyst: 10 mg — Example 7 - Example 7 chlorobenzene: 0.3 ml — Example 7 - Example 7 CDCl₃: 0.2 ml — Example 7 - Example 7 diethyl diallylmalonate: 0.15 ml — Example 7 - Example 7 time points: 30 min, 60 min — Example 7 - Example 8 CsBr: 1 mg — Example 8 - Example 8 Grubbs II catalyst: 10 mg — Example 8 - Example 8 chlorobenzene: 0.3 ml — Example 8 - Example 8 CDCl₃: 0.2 ml — Example 8 - Example 8 diethyl diallylmalonate: 0.15 ml — Example 8 - Example 8 time points: 15 min, 30 min, 60 min, 90 min, 150 min — Example 8 - Hydrogenation temperature: 60-150°C — Synthesis passage - Depolymerization temperature: 30-100°C — Synthesis passage

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis relative to the no-salt control. - Example 8: CsBr also accelerated ring-closing metathesis, though less strongly than LiBr in the reported table. - Synthesis passage: the patent states that metathesis degradation of NBR can be followed by hydrogenation to make low-Mooney HNBR. - Synthesis passage: the catalyst system is intended to improve metathesis efficiency and reduce catalyst loading in nitrile rubber degradation.

Technical Relevance This patent is highly relevant because it provides the metathesis catalyst platform used to lower NBR molecular weight before hydrogenation. That strategy directly addresses one of the key manufacturing challenges for HNBR: obtaining low-Mooney, processable feedstocks without excessive degradation.

Patent 4

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low molecular weight hydrogenated nitrile rubber made by first metathesis-degrading nitrile rubber with a co-olefin and ruthenium-based catalyst, then hydrogenating the degraded polymer. The process is designed to reduce Mooney viscosity and narrow molecular weight distribution while retaining the HNBR microstructure. - Rubber cement concentration: 15 wt.-% solid — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber cement concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-Hexene: 150 g — Example 4 - Dry N₂ degassing: 3 times — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst

load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 summary - Mw: 54 kg/mol — Table 2 summary - Mooney viscosity: 43 — Table 2 summary - Mn: 60 kg/mol — Table 2 summary - Mw: 136 kg/mol — Table 2 summary - Mooney viscosity: 28 — Table 2 summary - Mn: 31 kg/mol — Table 2 summary - Mw: 59 kg/mol — Table 2 summary - Mooney viscosity: 3 — Table 2 summary - Mn: 55 kg/mol — Table 2 summary - Mw: 111 kg/mol — Table 2 summary - Mooney viscosity: 31 — Table 2 summary

Relevant Experimental Evidence - Example 1: ethylene-assisted metathesis at 80°C and 0.05 phr catalyst reduced molecular weight to Mn 27 kg/mol and Mw 54 kg/mol. - Example 2: increasing catalyst load to 0.10 phr further reduced Mooney viscosity to 28. - Example 3: lower solids concentration and 0.25 phr catalyst gave the lowest reported Mooney viscosity of 3. - Example 4: 1-hexene was used as co-olefin, giving Mw 111 kg/mol and Mooney viscosity 31. - Table 2: the metathesized products showed up to a fivefold reduction in Mw and PDI as low as 1.90.

Technical Relevance This patent is directly relevant because it is one of the clearest disclosures of metathesis-assisted molecular weight reduction for HNBR precursors. It provides concrete process parameters and property outcomes that are useful for designing low-viscosity HNBR feedstocks prior to hydrogenation.

Patent 5

- Patent Number: EP3331072B1
- Patent Title: Composition for forming secondary battery cathode, and secondary battery ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a direct HNBR synthesis disclosure; it is a battery electrode formulation patent that uses a partially hydrogenated nitrile rubber as a dispersing agent for conductive carbon. The rubber is incorporated into a conductive material-dispersing agent composite prepared by bead milling, then combined with active material and binder to make a positive electrode slurry. - Rotor diameter: 150 mm — Example 1 - Rotor speed: 1,000 rpm — Example 1 - Rotor peripheral speed: 7.9 m/s — Example 1 - Bead diameter: 1.25 mm — Example 1 - Bead filling ratio: 80 wt% — Example 1 - Ejection speed: 1 kg/min — Example 1 - Carbon black specific surface area: 135 m²/g — Example 1 - Carbon black oil absorption number: 220 ml/100g — Claim 1 / Example 1 - Partially hydrogenated nitrile rubber hydrogenated conjugated diene repeat units: 20 to 50 wt% — Claim 1 - D50: 0.8 µm to 1.2 µm — Claim 1 - D90: 2.0 µm to 5.0 µm — Claim 2 - Ejection speed: 0.5 kg/min — Example 2 - Ejection speed: 1.2 kg/min — Example 3 - Temperature: 25 °C — Experimental Example 4 - Temperature: 45 °C — Experimental Example 4 - Charge/discharge rate: 1 C — Experimental Example 4 - Charge/discharge rate: 0.1 C — Experimental Example 3 - Charge/discharge rate: 2.0

Relevant Experimental Evidence - Example 1: a conductive material dispersed liquid was prepared using partially hydrogenated nitrile rubber and bead milling at 1,000 rpm. - Example 2: changing ejection speed to 0.5 kg/min altered particle size and battery performance metrics. - Example 3: changing ejection speed to 1.2 kg/min provided another dispersibility condition for comparison. - Experimental Example 2: resistance was measured at SOC15, SOC30, and SOC50 after cycling. - Experimental Example 4: high-temperature capacity retention was evaluated at 45°C over 490 cycles.

Technical Relevance This patent is relevant because it specifies a partially hydrogenated nitrile rubber grade and its hydrogenated diene content range, which helps define acceptable HNBR-like materials in electrochemical formulations. It also shows how hydrogenated nitrile rubbers are used when dispersion stability and mechanical integrity are required.

Patent 6

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions of nitrile rubber using ruthenium- or osmium-based carbene catalysts, with the goal of degrading NBR before hydrogenation to HNBR. The process is intended to lower molecular weight and improve processability while preserving the nitrile rubber backbone enough for subsequent hydrogenation. - Degree of hydrogenation: 50 to 100% — Detailed description - Mooney viscosity: 55 to 120 — Detailed description - Number average molecular weight: about 200000 to 700000 — Detailed description - Residual double bond content: 1 to 18% — Detailed description - Mooney viscosity of NBR feedstock: about 30 — Detailed description - Mooney viscosity after hydrogenation: about 55 — Detailed description - Mooney viscosity before hydrogenation: less than 30 — Detailed description - Number average molecular weight before hydrogenation: $M_n < 70000$ g/mol — Detailed description - Hydrogenated nitrile rubber Mw: 30000 to 250000 — Detailed description - Hydrogenated nitrile rubber Mooney viscosity: 3 to 50 — Detailed description - Hydrogenated nitrile rubber PDI: less than 2.5 — Detailed description - Metathesis degradation temperature: 30-100°C — synthesis passage - Hydrogenation temperature: 60-150°C — synthesis passage

Relevant Experimental Evidence - No examples were detected by the parser, but the detailed description states that NBR is metathesis-degraded before hydrogenation to produce low-Mooney

HNBR. - The patent contrasts Grubbs I and Grubbs II catalyst behavior for NBR metathesis and notes improved molecular weight control with Grubbs II. - The disclosure emphasizes that HNBR with Mooney viscosity below 55 and Mn below 200,000 g/mol was difficult to obtain by conventional routes. - The patent identifies the practical target as low-molecular-weight HNBR for improved processability.

Technical Relevance This patent is highly relevant because it explicitly addresses the core manufacturing problem for HNBR: reducing molecular weight before hydrogenation to obtain a processable product. It provides a catalyst-based route that complements conventional hydrogenation-only approaches.

Patent 7

- Patent Number: US20180022951A1
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a formulation patent for electroconductive silver paste that uses nitrile group-containing rubber, including HNBR, as a binder/dispersing matrix. It does not disclose HNBR synthesis, but it does specify that the rubber may be NBR and/or HNBR and that the HNBR-containing paste is prepared by dissolving the rubber, mixing with silver powder, and kneading on a three-roll mill. - Ratio by weight (A)/(B): 77/23 to 90/10 — Claim 1 - Alkali metal content: 4,000 ppm or less — Claim 1 - Alkali metal content: 1,500 ppm or less — Claim 3 - Nitrile group content: 25% by weight or more — Claim 2 - Specific resistance: less than $1.0 \times 10^{-3} \Omega \cdot \text{cm}$ — Claim 5 - Elongation at break: more than 35% — Claim 6 - Repetitive stretchability: 20% elongation, 50 times or more — Claim 6 - NBR nitrile group content: 51% by weight — Example 1/3 - NBR alkali metal content: 50 ppm — Example 1/3 - NBR alkali metal content: 1400 ppm — Example 3 - NBR alkali metal content: 5500 ppm — Example 3 - NBR nitrile group content: 36% by weight — Example 3/5 - NBR nitrile group content: 26% by weight — Example 3/6 - HNBR nitrile group content: 50% by weight — Example 3 - HNBR alkali metal content: 55 ppm — Example 3 - HNBR iodine value: 25 mg/100 mg — Example 3 - Specific resistance at 0% elongation: $1.5 \times 10^{-4} \Omega \cdot \text{cm}$ — Example 3 - Specific resistance at 100% elongation: $15 \times 10^{-4} \Omega \cdot \text{cm}$ — Example 3 - Anti-migration property: 200 seconds — Example 3 - Storage stability: 25°C for 3 months — Example 3 - Silver powder ratio (A)/(B): 81.5/18.5 — Example 1 - Silver powder ratio (A)/(B): 77.4/22.6 — Example 2 - Silver powder ratio (A)/(B): 89.4/10.6 — Example 3

Relevant Experimental Evidence - Preparation Example 1: NBR was polymerized from butadiene and acrylonitrile, then hydrogenated to HNBR (1). - Example 1: an NBR-based paste

with 81.5/18.5 silver/rubber ratio showed low initial resistance and good stretchability. - Example 2: increasing rubber fraction to 22.6% improved anti-migration but raised initial resistance slightly. - Example 3: HNBR (50 wt% nitrile group, 55 ppm alkali metal) was used and gave very good electrical and storage properties. - Example 5: a higher silver fraction caused breakage under elongation, showing the formulation sensitivity.

Technical Relevance This patent is relevant because it explicitly includes HNBR as a usable nitrile rubber binder and gives property windows for HNBR-containing formulations. It is useful for understanding how hydrogenation level and residual ionic content affect downstream processing and performance.

Patent 8

- Patent Number: EP1813835B1
- Patent Title: Toothed power transmission belt with cloth component thereon
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses toothed power transmission belts whose rubber body can be based on hydrogenated nitrile rubber, often combined with unsaturated carboxylic acid metal salts, silica, short fibers, and peroxide curing. The HNBR is used as a high-heat-resistance elastomer in the belt body, while the cloth component is engineered for adhesion and low friction. - Degree of hydrogenation: at least 90% — synthesis passage - Degree of hydrogenation: 92-98% — synthesis passage - Unsaturated carboxylic acid metal salt: 15-40 parts by mass to 100 parts polymer — synthesis passage - Silica: 10-60 parts by mass to 100 parts polymer — synthesis passage - Short fibers: 1-20 parts by mass per 100 parts polymer — synthesis passage - Viscosity increase during unvulcanization: may not increase — synthesis passage - Pressure vulcanization temperature: 165°C — Example section - Pressure vulcanization time: 30 minutes — Example section - Belt width: 30 mm — Comparative Example 8 - Tooth pitch: 9.525 mm — Comparative Example 8 - Driving pulley speed: 4,000 rpm — Comparative Example 8 - Tension: 3700 N — Comparative Example 8 - Initial tension: 350 N — Comparative Example 8 - Biaxial high load test load: 490 N·m — Comparative Example 5 - Driving pulley teeth: 33 — Comparative Example 5 - Driven pulley teeth: 61 — Comparative Example 5 - Driving pulley speed: 1200 rpm — Comparative Example 5 - Initial belt tension: 550 N — Comparative Example 5 - Inventive Example 10 life: 465 hours — Comparative Example 5

Relevant Experimental Evidence - Comparative Example A tables show multiple H-NBR and H-NBR-carboxylic acid formulations with varying carbon black, silica, TAIC, and TAC levels. - Example section: belts were pressure vulcanized at 165°C for 30 minutes and then tested for adhesion and friction. - Comparative Example 8: a belt with 120 ZBS30 geometry was run at 4,000

rpm and 3700 N tension for 200 hours. - Comparative Example 5: use of N,N'-m-phenylenebismaleimide led to excessive uncrosslinked viscosity and poor tooth formation. - The discussion concludes that HNBR plus unsaturated carboxylic acid metal salt and silica improves hardness and adhesion while maintaining processability.

Technical Relevance This patent is relevant because it uses highly hydrogenated nitrile rubber as the belt elastomer and defines practical hydrogenation and compounding windows. It shows how HNBR is formulated for heat-resistant, mechanically durable rubber articles.

Patent 9

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses novel ruthenium carbene complexes and their use in metathesis reactions, including depolymerization of NBR and hydrogenation of the depolymerized product to HNBR. The key synthesis route is not HNBR hydrogenation itself, but catalyst preparation and metathesis-enabled precursor modification that can be followed by hydrogenation at elevated temperature and pressure. - Ligand synthesis scale: 5.0 mmol — Example 1 - Reaction temperature: 85°C — Example 1 - Pd(PPh₃)₄: 2% — Example 1 - CuCl: 270 mg, 2.5 mmol, 2.5 eq — Example 1 - Ligand 3a yield: 86% — Example 1 - Ru complex 4a yield: 4% — Example 1 - Ligand synthesis scale: 5.0 mmol — Example 2 - CuCl: 270 mg, 2.5 mmol, 2.5 eq — Example 2 - Ru complex 4b yield: 29% — Example 2 - Ligand synthesis scale: 5.0 mmol — Example 3 - Ru complex 4c yield: 5% — Example 3 - Ligand synthesis scale: 5.0 mmol — Example 4 - Ru complex 4d yield: 32% — Example 4 - Ru complex 4e yield: 35% — Example 5 - Ru complex 4f yield: 14% — Example 6 - Ligand synthesis scale: 5.0 mmol — Example 7 - Ru complex 4g yield: 8% — Example 7 - Ligand synthesis scale: 5.0 mmol — Example 8 - Ru complex 4h yield: 42% — Example 8 - Ligand synthesis scale: 5.0 mmol — Example 9 - Ru complex 4j yield: 48% — Example 9 - Ligand synthesis scale: 5.0 mmol — Example 10 - Ru complex 4k yield: 68% — Example 10 - Ligand synthesis scale: 5.0 mmol — Example 11 - Ru complex 4m yield: 52% — Example 11 - Ligand synthesis scale: 5.0 mmol — Example 12 - Ru complex 4n yield: 64% — Example 12 - Ligand synthesis scale: 5.0 mmol — Example 13 - Ru complex 4p yield: 26% — Example 13 - Ligand synthesis scale: 5.0 mmol — Example 14 - Ru complex 4q yield: 38% — Example 14 - Ligand synthesis scale: 5.0 mmol — Example 15 - Ru complex 4r yield: 14% — Example 15 - Ligand synthesis scale: 5.0 mmol — Example 16 - Ru complex 4s yield: 85% — Example 16 - Ligand synthesis scale: 5.0 mmol — Example 17 - Ru complex 4t yield: 23% — Example 17 - Ligand synthesis scale: 5.0 mmol — Example 18 - Ru complex 4u yield: 32% — Example 18 - Ligand synthesis scale: 5.0 mmol —

Example 19 - Ru complex 4v yield: 10% — Example 19 - Ligand synthesis scale: 5.0 mmol —
Example 20 - Ru complex 4w yield: 31% — Example 20 - Depolymerization temperature:
30-100°C — synthesis passage - Hydrogenation temperature: 60-150°C — synthesis passage

Relevant Experimental Evidence - Example 1: ligand 3a was prepared in 86% yield and converted to Ru complex 4a in 4% yield. - Example 2: ligand 3b gave Ru complex 4b in 29% yield. - Example 8: ligand 3h gave Ru complex 4h in 42% yield. - Example 16: ligand 3s gave Ru complex 4s in 85% yield, one of the highest reported complex yields. - Example 20: ligand 3w was converted to Ru complex 4w in 31% yield using the alternative Ru reagent.

Technical Relevance This patent is relevant because it provides the catalyst chemistry enabling NBR depolymerization prior to hydrogenation. Such metathesis-based precursor modification is a key route to low-Mooney HNBR with improved processability.

Patent 10

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a rubber compounding and vulcanization-molding disclosure in which hydrogenated nitrile rubber is listed as one possible base rubber among many. The process centers on blending hollow particles with a base rubber, then kneading and vulcanization molding to obtain lightweight molded rubber products with good dimensional stability. - Base rubber Mooney viscosity: 54 — Example 1 - Hollow particles further expansion ratio: 50.0% — Example 1 - Hollow particles amount: 2.05 parts by weight — Example 1 - Kneading temperature: 60°C — Example 1 - Vulcanization molding temperature: 140°C — Example 1 - Vulcanization molding time: 30 minutes — Example 1 - Hollow particles further expansion ratio: 22.2% — Table 4 - Hollow particles amount: 0.9 parts by weight — Table 4 - Hollow particles amount: 2.7 parts by weight — Table 4 - Hollow particles amount: 6.2 parts by weight — Table 4 - Hollow particles amount: 9.0 parts by weight — Table 4 - Base rubber Mooney viscosity: 30 — Table 4 - Kneading method: Open roll mill — Table 4 - Vulcanization molding temperature: 150°C — Table 4 - Vulcanization molding temperature: 170°C — Table 4 - Vulcanization molding temperature: 180°C — Table 4 - Injection molding cylinder temperature: 180°C — Example 6 - Injection molding die temperature: 60°C — Example 6 - Injection pressure: 15 MPa — Example 6 - Hollow particles mean particle size: 129.8 μm — Example 11 - Further expansion ratio: 20 to 80% — Claim 1 - Base rubber Mooney viscosity: 5 to 90 at 100°C — Claim 1 - Amount of hollow particles: 0.1 to 30 wt% — Claim 2 - Amount of base rubber: 10 to 95 wt% — Claim 2

Relevant Experimental Evidence - Example 1: hollow particles with 50% further expansion ratio in an EPDM base rubber gave excellent lightweight effect and dimensional stability. - Example 3: hollow particles with 22.2% further expansion ratio were extruded and vulcanized at 150°C for 15 minutes with good appearance. - Example 6: injection molding at 180°C cylinder temperature and 15 MPa produced a lightweight molded product with good dimensional stability. - Example 11: large hollow particles with mean size 129.8 μm still gave acceptable products without re-expansion during vulcanization. - Comparative examples show that too-low or too-high further expansion ratios caused breakage or re-expansion problems.

Technical Relevance This patent is only indirectly relevant to HNBR because hydrogenated nitrile rubber is one of the listed base rubbers for the molded composition. It is useful for understanding how HNBR can be processed in lightweight vulcanized articles, but it does not disclose HNBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Two patents directly address HNBR precursor modification by metathesis before hydrogenation: US7951875B2 and US7737233B2. Both use ruthenium-based metathesis chemistry to reduce molecular weight, but US7951875B2 provides more concrete operating data (80°C, 0.05-0.25 phr catalyst, ethylene or 1-hexene co-olefin), while US7737233B2 emphasizes catalyst system design and broader process windows.
- Direct hydrogenation disclosures in EP2692788B1 use latex or solution hydrogenation with palladium catalysts at 3.0-3 MPa and 50°C, whereas the metathesis patents focus on pre-hydrogenation chain scission to lower Mooney viscosity.
- EP2692788B1 and EP1813835B1 both use highly saturated nitrile rubber in formulations, but EP2692788B1 targets polyamide-containing crosslinkable compositions and EP1813835B1 targets toothed belts with unsaturated carboxylic acid metal salts, silica, and short fibers.
- US10385647B2, US20180022951A1, EP3331072B1, EP1813835B1, and US10731020B2 are formulation patents rather than synthesis patents; they define useful HNBR property windows such as hydrogenation level, alkali metal content, and thermal stability in end-use systems.
- US20180022951A1 specifies HNBR with 50 wt% nitrile group and 25 mg/100 mg iodine value in a silver paste, while EP3331072B1 specifies partially hydrogenated nitrile rubber with 20-50 wt% hydrogenated conjugated diene repeat units for conductive dispersion.
- US10731020B2 is the least synthesis-relevant to HNBR, but it confirms that hydrogenated nitrile rubber is acceptable as a base rubber in vulcanization-molded lightweight articles alongside EPDM, NBR, and other elastomers.

4. CONCLUSION

The patent evidence supports two practical HNBR synthesis strategies: direct hydrogenation of NBR precursors in latex or solution, and metathesis-assisted molecular weight reduction followed by hydrogenation. For direct hydrogenation, the clearest disclosed conditions are palladium-catalyzed hydrogenation at about 3.0-3 MPa and 50°C, with latex coagulation and drying after hydrogenation. For low-Mooney, processable HNBR, the strongest evidence favors ruthenium-catalyzed metathesis of NBR with a co-olefin, using catalyst loadings in the 0.05-0.25 phr range and temperatures around 80°C, followed by hydrogenation at 60-150°C. Across the formulation patents, useful HNBR grades are those with high hydrogenation, low residual unsaturation, low alkali metal content, and sufficient thermal stability for brine, battery, belt, and conductive paste applications. The recommended synthesis direction from the evidence is to control NBR feedstock composition first, then reduce molecular weight by metathesis if low Mooney viscosity is needed, and finally hydrogenate under moderate pressure and temperature to preserve molecular weight distribution and achieve the desired HNBR performance.

5. REFERENCES

- EP2692788B1 | Highly saturated nitrile rubber composition and crosslinked rubber | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2692788B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP3331072B1 | Composition for forming secondary battery cathode, and secondary battery ... | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3331072B1/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US20180022951A1 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180022951A1/en>
- EP1813835B1 | Toothed power transmission belt with cloth component thereon | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP1813835B1/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>